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AQEval: R code for the analysis of discrete change in Air Quality time-series

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Software

- Review
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Summary

`AQEval` (Air Quality Evaluation) is an `R` package of functions for the routine investigation of discrete changes in air quality time-series, e.g. assessing the impact of interventions and/or pollution events. The main functions use break-point/segmentation (BP/S) methods to detect, characterise and quantify change, while other functions build on these to provide a workflow for those looking to measure smaller changes and/or changes in more complex environments.

Statement of Need

Authorities responsible for air quality management are typically required to implement and evaluate the air quality interventions they adopt (Bradley et al., 2019). These interventions are often costly, disruptive and unpopular (Glazener & Khreis, 2019), and associated benefits are often hard to measure because of the inherent variability of air quality data (Grange & Carslaw, 2019; Jones et al., 2012; Kelly et al., 2011; Pearce et al., 2011).

Various methods have been developed to investigate discrete changes in a wide range of time-series (see e.g. Reeves et al., 2007; Truong et al., 2020), and several `R` (R Core Team & others, 2025) packages have been developed for their use, e.g. `bcp` (Erdman & Emerson, 2008), `changepoint` (Killick et al., 2016), `segmented` (Muggeo & others, 2008), and `strucchange` (Zeileis et al., 2002). Some have even been applied to the detection of large changes in air quality time-series (Carslaw et al., 2006), and with additional signal isolation to smaller changes (Carslaw & Carslaw, 2007). However, those tasked with air quality data analysis, although highly skilled in a wide range of air quality monitoring and assessment activities, are unlike to be able to dedicate sufficient time and funding to the development of in-house expertise required to routinely apply these.

`AQEval` was developed to address this skill gap. It aligns the inputs and outputs of a number of statistical methods to provide a one-package option for anyone interested in using `R` to routinely detect, characterise and quantify discrete change in air quality data.

As many air quality professionals already use the `R` package `openair` (Carslaw & Ropkins, 2012; Ropkins & Carslaw, 2012) for more conventional analysis and data visualisation, `AQEval` has also been written using `openair` coding conventions and function structure to reduce the learning-curve typical associated with learning new software.

Sources

AQEval is freely available under General Public License (GPL) from conventional on-line R archives:

- The latest (stable) release version of AQEval is on the Comprehensive R Archive Network (CRAN) at https://CRAN.R-project.org/package=AQEval; and,
- The developers' version and code are publicly available on GitHub at https://github.com/karlropkins/AQEval, which can also be used to report issues or suggest changes.
- The project website is at https://karlropkins.github.io/AQEval/.

Analytical Rationale

The AQEval Break-Point/Segment (BP/S) methods involve three steps: finding possible 'points of change', using break-points, and then characterising and quantifying 'regions of change' about these as segments:

1. Breaks-points are found using the strucchange methods of Zeileis and colleagues (Zeileis et al., 2002, 2003). Here, a rolling-window approach is applied: a first subset of data (or time-series window, TW_0 in Figure 1a), is selected and a linear regression model built; the window is then rolled on, typically one measurement (TW_1 in Figure 1a) and a second model build; and so on, until the end of the time-series; then, points of likely change are identified using the F-Stat measure of difference for sequential models.
2. In addition to the standard Bayesian Information Criterion (BIC) testing used by strucchange, AQEval also checks that all terms associated which each break-point are statistically valid ($p<0.05$), discarding any combinations generating invalid models. This approach was selected on the basis of simulation testing.
3. Finally, the segmented methods of Muggeo and colleagues (Muggeo, 2003, 2017; Muggeo & others, 2008) are used to extend the break-point to break-segments. Here, the confidence intervals for the selected break-points are used as the start points when building the segmented model, and final segment ranges are assigned based on random walk testing the regions about these points as illustrated in Figure 1b.

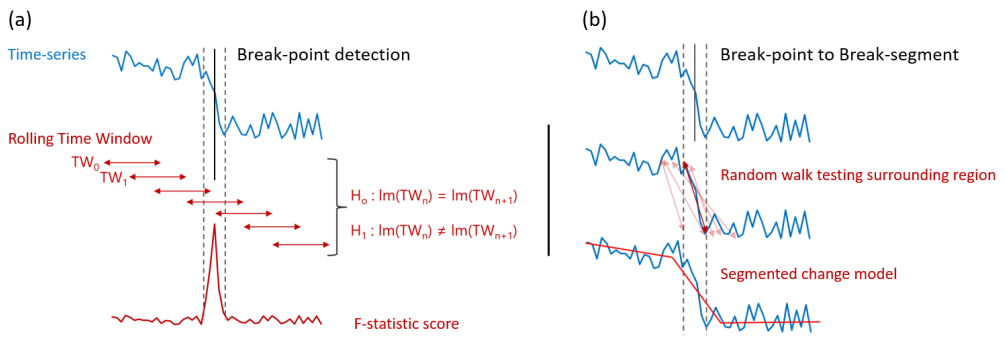

Figure 1: The basic break-point/segment scheme: (a) Break-point to identified change-points, and (b) segment modelling of the regions about the break-points to produce break-segments.

Figure 2 shows the break-point/segment analysis of an NO_2 time-series from a heavily trafficked site in central London where a change event (*ca.* 25 μg.m^{-3}; 31%) is detected between 2003-01-11 and 2003-02-19, and detected using standard break-point/segment analysis.

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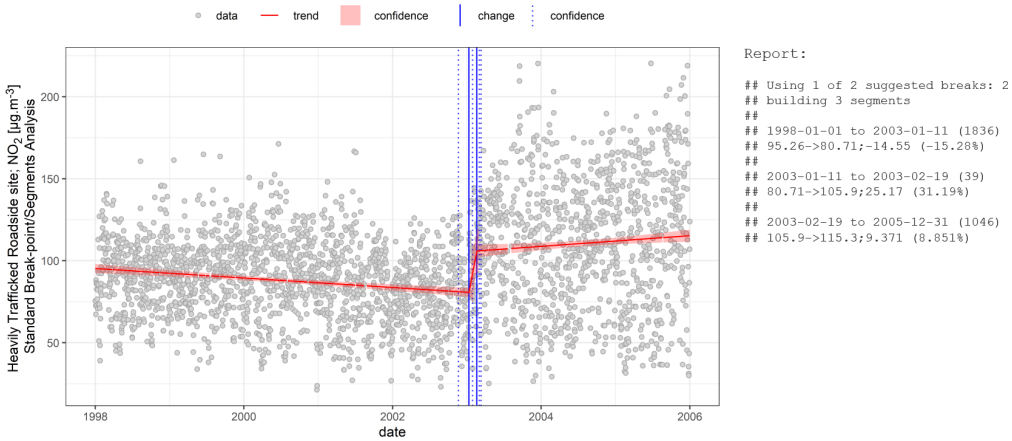

Figure 2: Standard AQEval break-point/segment analysis (graphical output and report) of NO_2 1998-2005 time-series analysis from Marylebone Road, a heavily trafficked roadside in the UK.

In some cases changes are small or local air inputs are complex, and time-series may require additional pre-processing to successfully isolate an obscured change-event. For example, if a studied time-series is several years long, a small change may be hidden by seasonally variability in meteorological contributions. `AQEval` includes Generalized Additive Model (GAM) based methods (based on `mgcv`, Wood, 2017, 2025) to build and subtract such variance. By default, these take the form:

$$[pollutant] = s_1(day\text{-}of\text{-}year) + s_2(hour\text{-}of\text{-}day) + te_1(wind\ speed,\ wind\ direction)$$

$$[pollutant]_{isolated} = ([pollutant] - [pollutant]_{predicted}) + mean(pollutant)$$

Where the investigate pollutant concentration, *[pollutant]*, is modelled as a function of day-of-year, hour-of-day and wind speed and direction using a combination spline (s_1 and s_2) and tensor (te_1) fit-terms, and the unmodelled component, $[pollutant]_{isolated}$, is estimated as the mean-centred residual of this model prediction.

This default correction can also be modified to include other potential confounders, e.g. other frequency terms (e.g. day-of-week and/or week-of-year), background contributions (by subtraction of local variance associated with trends at near-by site not affected by the investigated change), or proxies for other local contributors (e.g. other meteorological parameters like air temperature, markers for other sources, etc). Figure 3a shows the break-point analysis of NO_2 from a nearby but less heavily trafficked site where seasonality dominates the time-series, and Figure 3b shows the smaller (*ca.* 6.6 $\mu g.m^{-3}$; 13%) underlying change-event observed at a similar time to the large change observed at the more heavily trafficked site in Figure 2 (2002-09-09 to 2002-12-21 compared with 2003-01-11 and 2003-02-19).

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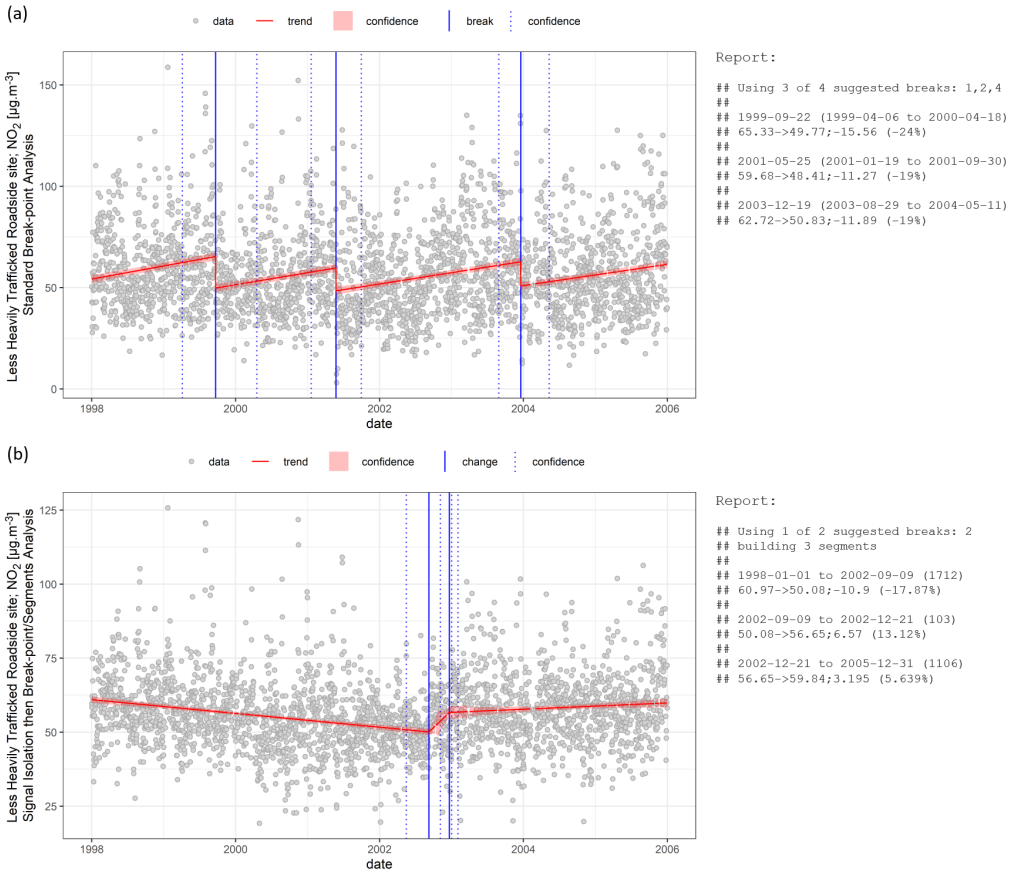

Figure 3: AQEval analysis of NO_2 1998-2005 time-series at Ealing Acton Town Hall, a less heavily trafficked roadside site near Marylebone Road: (a) the standard break-point analysis of the ambient time-series exhibits a near-regular distribution of breaks typical of a site dominated by seasonal factors; and, (b) shows the underlying change-event revealed using signal isolation and then break-point/segment analysis.

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Related Outputs

Detailed descriptions of the `AQEval` functions, and an extended discussion and analysis of the examples used in Figures 2 and 3 is provided in the extended package introduction.
Other work using `AQEval` include:

- Ropkins & Tate (2021), a peer-reviewed article reporting on a multi-species AQEval analysis of the air quality impact of the COVID-19 lockdown in the UK.
- Ropkins et al (2022), a peer-reviewed article reporting on the use of `AQEval` to measure the impact of a bus fleet intervention on NO_2 levels.
- Also Clear Air Zone (CAZ) impact assessment reports include analyses using `AQEval`, see e.g.:
  - CAZ Baseline Study
  - First Year Report
  - (...)
  - Report archive

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The views and opinions expressed herein by the authors are their own and do not necessarily reflect those of UK Government or any agency thereof.

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